

Aggregation Stages

In the `db.collection.aggregate()` method and `db.aggregate()` method, pipeline stages appear in an array. In the Atlas UI, you can arrange pipeline stages using the aggregation pipeline builder. Documents pass through the stages in sequence.

Compatibility

You can use pipeline stages for deployments hosted in the following environments:

- MongoDB Atlas: The fully managed service for MongoDB deployments in the cloud
- MongoDB Enterprise: The subscription-based, self-managed version of MongoDB
- MongoDB Community: The source-available, free-to-use, and self-managed version of MongoDB

db.collection.aggregate() Stages

All stages except the `$out`, `$merge`, `$geoNear`, `$changeStream`, and `$changeStreamSplitLargeEvent` stages can appear multiple times in a pipeline.

NOTE

For details on a specific operator, including syntax and examples, click on the link to the operator's reference page.

```
db.collection.aggregate( [ { <stage> }, ... ] )
```



Stage	Description
<code>\$addFields</code>	Adds new fields to documents. Similar to <code>\$project</code> , <code>\$addFields</code> reshapes each document in the stream; specifically, by adding new fields to output documents that contain both the existing fields from the input documents and the newly added fields. <code>\$set</code> is an alias for <code>\$addFields</code> .
<code>\$bucket</code>	Categorizes incoming documents into groups, called buckets, based on a specified expression and bucket boundaries.
<code>\$bucketAuto</code>	Categorizes incoming documents into a specific number of groups, called buckets, based on a specified expression. Bucket boundaries are automatically determined in an attempt to evenly distribute the documents into the specified number of buckets.
<code>\$changeStream</code>	Returns a Change Stream cursor for the collection. This stage can only occur once in an aggregation pipeline and it must occur as the first stage.
<code>\$changeStreamSplitLargeEvent</code>	Splits large change stream events that exceed 16 MB into smaller fragments returned in a change stream cursor.

Stage	Description
	You can only use <code>\$changeStreamSplitLargeEvent</code> in a <code>\$changeStream</code> pipeline and it must be the final stage in the pipeline.
<code>\$collStats</code>	Returns statistics regarding a collection or view.
<code>\$count</code>	Returns a count of the number of documents at this stage of the aggregation pipeline. Distinct from the <code>\$count</code> aggregation accumulator.
<code>\$densify</code>	Creates new documents in a sequence of documents where certain values in a field are missing.
<code>\$documents</code>	Returns literal documents from input expressions.
<code>\$facet</code>	Processes multiple aggregation pipelines within a single stage on the same set of input documents. Enables the creation of multi-faceted aggregations capable of characterizing data across multiple dimensions, or facets, in a single stage.
<code>\$fill</code>	Populates <code>null</code> and missing field values within documents.
<code>\$geoNear</code>	Returns an ordered stream of documents based on the proximity to a geospatial point. Incorporates the functionality of <code>\$match</code> , <code>\$sort</code> , and <code>\$limit</code> for geospatial data. The output documents include an additional distance field and can include a location identifier field.
<code>\$graphLookup</code>	Performs a recursive search on a collection. To each output document, adds a new array field that contains the traversal results of the recursive search for that document.
<code>\$group</code>	Groups input documents by a specified identifier expression and applies the accumulator expression(s), if specified, to each group. Consumes all input documents and outputs one document per each distinct group. The output documents only contain the identifier field and, if specified, accumulated fields.
<code>\$indexStats</code>	Returns statistics regarding the use of each index for the collection.
<code>\$limit</code>	Passes the first <i>n</i> documents unmodified to the pipeline where <i>n</i> is the specified limit. For each input document, outputs either one document (for the first <i>n</i> documents) or zero documents (after the first <i>n</i> documents).
<code>\$listSampledQueries</code>	Lists sampled queries for all collections or a specific collection.
<code>\$listSearchIndexes</code>	Returns information about existing Atlas Search indexes on a specified collection.
<code>\$listSessions</code>	Lists all sessions that have been active long enough to propagate to the <code>system.sessions</code> collection.
<code>\$lookup</code>	Performs a left outer join to another collection in the <i>same</i> database to filter in documents from the "joined" collection for processing.
<code>\$match</code>	Filters the document stream to allow only matching documents to pass unmodified into the next pipeline stage. <code>\$match</code> uses standard MongoDB queries. For each input document, outputs either one document (a match) or zero documents (no match).
<code>\$merge</code>	Writes the resulting documents of the aggregation pipeline to a collection. The stage can incorporate (insert new documents, merge documents, replace documents, keep existing

Stage	Description
	documents, fail the operation, process documents with a custom update pipeline) the results into an output collection. To use the <code>\$merge</code> stage, it must be the last stage in the pipeline.
<code>\$out</code>	Writes the resulting documents of the aggregation pipeline to a collection. To use the <code>\$out</code> stage, it must be the last stage in the pipeline.
<code>\$planCacheStats</code>	Returns plan cache information for a collection.
<code>\$project</code>	Reshapes each document in the stream, such as by adding new fields or removing existing fields. For each input document, outputs one document. See also <code>\$unset</code> for removing existing fields.
<code>\$queryStats</code>	Returns runtime statistics for recorded queries. WARNING The <code>\$queryStats</code> aggregation stage is unsupported and is not guaranteed to be stable in a future release. Don't build functionality that relies on a specific output form of this stage, since the output may change in a future release.
<code>\$redact</code>	Reshapes each document in the stream by restricting the content for each document based on information stored in the documents themselves. Incorporates the functionality of <code>\$project</code> and <code>\$match</code> . Can be used to implement field level redaction. For each input document, outputs either one or zero documents.
<code>\$replaceRoot</code>	Replaces a document with the specified embedded document. The operation replaces existing fields in the input document, including the <code>_id</code> field. Specify a document embedded in the input document to promote the embedded document to the top level. <code>\$replaceWith</code> is an alias for <code>\$replaceRoot</code> stage.
<code>\$replaceWith</code>	Replaces a document with the specified embedded document. The operation replaces existing fields in the input document, including the <code>_id</code> field. Specify a document embedded in the input document to promote the embedded document to the top level. <code>\$replaceWith</code> is an alias for <code>\$replaceRoot</code> stage.
<code>\$sample</code>	Randomly selects the specified number of documents from its input.
<code>\$search</code>	Performs a full-text search of the field or fields in an Atlas collection. NOTE <code>\$search</code> is only available for MongoDB Atlas clusters, and is not available for self-managed deployments. To learn more, see Atlas Search Aggregation Pipeline Stages.
<code>\$searchMeta</code>	Returns different types of metadata result documents for the Atlas Search query against an Atlas collection. NOTE <code>\$searchMeta</code> is only available for MongoDB Atlas clusters running MongoDB v4.4.9 or higher, and is not available for self-managed deployments. To learn more, see Atlas Search Aggregation Pipeline Stages.

Stage	Description
<code>\$set</code>	Adds new fields to documents. Similar to <code>\$project</code>, <code>\$set</code> reshapes each document in the stream; specifically, by adding new fields to output documents that contain both the existing fields from the input documents and the newly added fields. <code>\$set</code> is an alias for <code>\$addField</code> stage.
<code>\$setWindowFields</code>	Groups documents into windows and applies one or more operators to the documents each window. <i>New in version 5.0.</i>
<code>\$skip</code>	Skips the first n documents where n is the specified skip number and passes the remaining documents unmodified to the pipeline. For each input document, outputs either zero documents (for the first n documents) or one document (if after the first n documents).
<code>\$sort</code>	Reorders the document stream by a specified sort key. Only the order changes; the documents remain unmodified. For each input document, outputs one document.
<code>\$sortByCount</code>	Groups incoming documents based on the value of a specified expression, then computes the count of documents in each distinct group.
<code>\$unionWith</code>	Performs a union of two collections; i.e. combines pipeline results from two collections into a single result set.
<code>\$unset</code>	Removes/excludes fields from documents. <code>\$unset</code> is an alias for <code>\$project</code> stage that removes fields.
<code>\$unwind</code>	Deconstructs an array field from the input documents to output a document for <i>each</i> element. Each output document replaces the array with an element value. For each input document, outputs n documents where n is the number of array elements and can be zero for an empty array.
<code>\$vectorSearch</code>	Performs an <u>ANN</u> search on a vector in the specified field of an Atlas collection. <i>New in version 7.0.2.</i> NOTE <code>\$vectorSearch</code> is only available for MongoDB Atlas clusters running MongoDB v6.0.11 or higher, and is not available for self-managed deployments. To learn more, see Atlas Search Aggregation Pipeline Stages.

For aggregation expression operators to use in the pipeline stages, see Aggregation Operators.

db.aggregate() Stages

MongoDB also provides the `db.aggregate()` method:

```
db.aggregate( [ { <stage> }, ... ] )
```



The following stages use the `db.aggregate()` method and not the `db.collection.aggregate()` method.

Stage	Description
<code>\$changeStream</code>	Returns a Change Stream cursor for the collection. This stage can only occur once in an aggregation pipeline and it must occur as the first stage.
<code>\$currentOp</code>	Returns information on active and/or dormant operations for the MongoDB deployment.
<code>\$documents</code>	Returns literal documents from input values.
<code>\$listLocalSessions</code>	Lists all active sessions recently in use on the currently connected <code>mongos</code> or <code>mongod</code> instance. These sessions may have not yet propagated to the <code>system.sessions</code> collection.

Stages Available for Updates

You can use the aggregation pipeline for updates in:

Command	<code>mongosh</code> Methods
<code>findAndModify</code>	<code>db.collection.findOneAndUpdate()</code> <code>db.collection.findAndModify()</code>
<code>update</code>	<code>db.collection.updateOne()</code> <code>db.collection.updateMany()</code> <code>Bulk.find.update()</code> <code>Bulk.find.updateOne()</code> <code>Bulk.find.upsert()</code>

For the updates, the pipeline can consist of the following stages:

- `$addField` and its alias `$set`
- `$project` and its alias `$unset`
- `$replaceRoot` and its alias `$replaceWith`.